LieRing Beam Physics Manual

canonical variables, Lie maps, RF, acceleration

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1 Beam Physics Manual

Symplectic modelling of a magnetic storage ring with an RF cavity. The independent variable is the arc length s along the reference orbit. All dynamics are written in **canonical** phase-space variables; laboratory angles and RF phase appear only as derived output.

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Figures use the Nuclotron-based lattice at deuteron kinetic energy 270 MeV unless a ramp from 10 MeV is shown. A separate *Programmer Manual* documents the implementation.

2 Canonical phase space

2.1 Variables

The dynamical state at arc length s is the six-vector

$$z = (x,\, p_x,\, y,\, p_y,\, l,\, \delta).$$

- x, y: transverse displacements from the reference orbit [m]. - $p_x = P_x/p_0$, $p_y = P_y/p_0$: canonical momenta normalised to the reference momentum p_0. - $\delta = (p - p_0)/p_0$: relative *momentum* deviation. - $l = -v(\delta)\,\Delta t$: longitudinal coordinate conjugate to δ, with $\Delta t = t - t_0(s)$ the time lag relative to the reference particle and $v(\delta)$ the particle speed,

$$\frac{v(\delta)}{v_0} = \frac{1+\delta}{\beta_0\sqrt{(1+\delta)^2 + 1/(\beta_0\gamma_0)^2}}.$$

The reference particle has charge Ze, rest energy mc^2, Lorentz factor γ_0, velocity $v_0 = \beta_0 c$ and momentum p_0.

The symplectic form on this space is the standard one,

$$\omega = \mathrm{d}x \wedge \mathrm{d}p_x + \mathrm{d}y \wedge \mathrm{d}p_y + \mathrm{d}l \wedge \mathrm{d}\delta,$$

or in matrix form $S = \mathrm{diag}(S_2, S_2, S_2)$ with $S_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. A map $z \mapsto \mathcal{M}(z)$ is symplectic iff its Jacobian satisfies $J^{\mathsf{T}} S J = S$.

2.2 Output (non-canonical) coordinates

For plots and comparison with laboratory data one uses

$$u = (x,\, x',\, y,\, y',\, \phi,\, \delta), \qquad x' = \frac{p_x}{p_s}, \quad p_s = \sqrt{(1+\delta)^2 - p_x^2 - p_y^2}, \quad \phi = -h\omega_0\Delta t = \frac{2\pi h}{C}\,\frac{l}{v(\delta)/v_0}.$$

$\phi > 0$ means early arrival. The set u is **not** canonical: x' depends on δ and p_y, and ϕ is scaled by $1/v(\delta)$. Symplecticity is therefore checked only in z. The conversion $z \leftrightarrow u$ is exact (nonlinear); truncated polynomial maps apply it as a full Taylor polynomial to the map order.

2.3 Energy term

Every magnetic Hamiltonian uses

$$e(\delta) = \frac{1}{\beta_0}\sqrt{(1+\delta)^2 + \frac{1}{\beta_0^2\gamma_0^2}} - \frac{1}{\beta_0^2},$$

chosen so that the irrelevant constant vanishes at $\delta = 0$ after removal of the constant term of H.

3 Element Hamiltonians

With $p_s = \sqrt{(1+\delta)^2 - p_x^2 - p_y^2}$, curvature $h = 1/\rho$ and normalised multipoles $k_1 = B'/(B\rho)$, $k_2 = B''/(B\rho)$, the s-Hamiltonians (hard-edge, no fringe fields) are

element	$H(z)$
Drift	$e(\delta) - p_s$
Quadrupole	$e(\delta) - p_s + \frac{k_1}{2}(x^2 - y^2)$
Sextupole	$e(\delta) - p_s + \frac{k_2}{6}(x^3 - 3xy^2)$
Sector bend	$e(\delta) - (1+hx)p_s + hx + \frac{1}{2}h^2x^2 + \frac{k_1}{2}(x^2 - y^2)$
Marker	0
Finite RF cavity	$e(\delta) - p_s + A\cos\Phi(l, \delta, s)$ (see Finite-length RF cavity)

For the bend, $\partial H/\partial x|_{z=0} = 0$: the reference arc is a solution of Hamilton's equations.

3.1 Maps from a Hamiltonian

An s-independent Hamiltonian over length L generates the Lie map

$$\mathcal{M} = \exp(-L : H :), \qquad : f : g = [f, g] = \sum_i \Big(\partial_{q_i} f \, \partial_{p_i} g - \partial_{p_i} f \, \partial_{q_i} g \Big).$$

Truncation of the Taylor series of $\mathcal{M}$ at a finite order yields a polynomial map that is symplectic only up to that order. Factorising

$$\mathcal{M} = R \circ e^{:f_3:} \circ e^{:f_4:}$$

(Dragt–Finn) and integrating each nonlinear factor by the implicit midpoint rule restores symplecticity to round-off. An independent construction integrates Hamilton's equations $\mathrm{d}z/\mathrm{d}s = S\nabla H$ by a Runge–Kutta scheme in the truncated polynomial algebra (differential-algebra map) and is used as a cross-check.

4 Linear optics

The linear map of an element is $R = \exp(L\,S\,\mathcal{H})$ with $\mathcal{H}_{ij} = \partial^2 H/\partial z_i \partial z_j|_0$. The one-turn matrix M is the product over the ring. With $\cos\mu = \frac{1}{2}\mathrm{Tr}M_x$,

$$\beta = \frac{M_{12}}{\sin\mu}, \qquad \alpha = \frac{M_{11} - M_{22}}{2\sin\mu},$$

and the closed dispersion solves $(I - M_4)D = M_{\cdot 6}$. Momentum compaction to first order:

$$\alpha_0 = \frac{1}{\gamma_0^2} - \frac{(l|\delta)_{\mathrm{closed}}}{C} = \frac{1}{C}\oint h\,D\,\mathrm{d}s.$$

Twiss and dispersion propagated in slices close after one turn to machine precision.

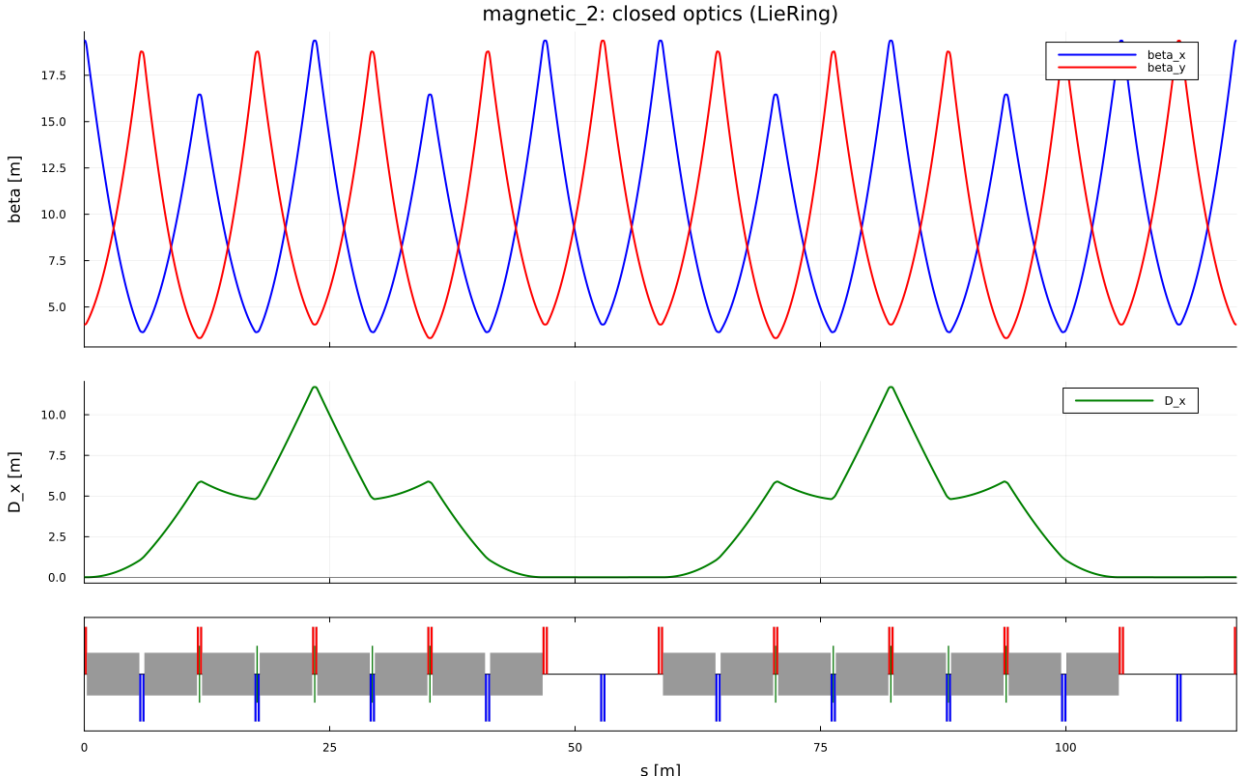


Figure 1: Closed beta functions, horizontal dispersion and magnet layout.

5 Second-order dispersion and momentum compaction

5.1 Closed orbit in δ

$$z_\perp(\delta) = D_1\delta + D_2\delta^2 + O(\delta^3),$$

from $M_\perp(z_\perp(\delta), \delta) = z_\perp(\delta)$ order by order.

5.2 Path length and α_i

On the closed orbit the geometric circumference admits the expansion

$$\frac{C(\delta)}{C_0} = 1 + \alpha_0\delta + \alpha_1\delta^2 + \alpha_2\delta^3 + \cdots .$$

With the canonical coordinate l related to time of flight by $l = -v\Delta t$, one turn gives $\Delta l = -C(\delta) + C_0\, v(\delta)/v_0$, hence

$$\alpha_0 = -\frac{l_1}{C} + \frac{1}{\gamma_0^2}, \qquad \alpha_1 = -\frac{l_2}{C} + c_2, \qquad \alpha_2 = -\frac{l_3}{C} + c_3,$$

where $v/v_0 - 1 = c_1\delta + c_2\delta^2 + c_3\delta^3$ and $c_1 = 1/\gamma_0^2$. Equivalently, α_i may be fitted from $C(\delta)/C_0$ evaluated on closed orbits at several δ.

5.3 Slip factor

The relative change of revolution period is $\Delta T/T_0 = \eta_0\delta + \eta_1\delta^2 + \eta_2\delta^3 + \cdots$. Directly from time of flight, $\eta_0 = -l_1/C = \alpha_0 - 1/\gamma_0^2$. The BLonD convention (Equations of Motion, eq. 15) expresses higher η_i through α_i:

$$\begin{aligned}
\eta_0 &= \alpha_0 - \frac{1}{\gamma_s^2}, \\
\eta_1 &= \frac{3\beta_s^2}{2\gamma_s^2} + \alpha_1 - \alpha_0\eta_0, \\
\eta_2 &= -\frac{\beta_s^2(5\beta_s^2 - 1)}{2\gamma_s^2} + \alpha_2 - 2\alpha_0\alpha_1 + \frac{\alpha_1}{\gamma_s^2} + \alpha_0^2\eta_0 - \frac{3\beta_s^2\alpha_0}{2\gamma_s^2}.
\end{aligned}$$

The two expansions of η_1 (TOF identity vs BLonD) differ by terms of order α_0^2; both are consistent when used with the matching drift formula (see below).

5.4 BLonD longitudinal drift (comparison)

BLonD updates arrival time after an energy kick by (eq. 20, solver `exact`)

$$\Delta t \leftarrow \Delta t + T_0\left[\left(1 + \alpha_0\delta + \alpha_1\delta^2 + \alpha_2\delta^3\right)\frac{1 + \Delta E/E_s}{1 + \delta} - 1\right],$$

with δ computed exactly from ΔE, and by the linearisation (eq. 19, solver `simple`)

$$\Delta t \leftarrow \Delta t + T_0\,\eta_0\,\frac{\Delta E}{\beta_s^2 E_s}.$$

Element-by-element integration of the ring Hamiltonians reproduces the `exact` formula at the level $\sim 10^{-6}$ relative in $\Delta\phi$ per turn when $\alpha_{0,1,2}$ are taken from the lattice; the `simple` formula deviates at the level of $\alpha_1\delta$ (a few 10^{-3} at $|\delta| = 10^{-3}$). Thus α_1 *is* the leading nonlinear longitudinal term in the BLonD sense; the ring model includes it automatically through the exact element Hamiltonians (to all orders in the exact integrator, to map order in the truncated maps).

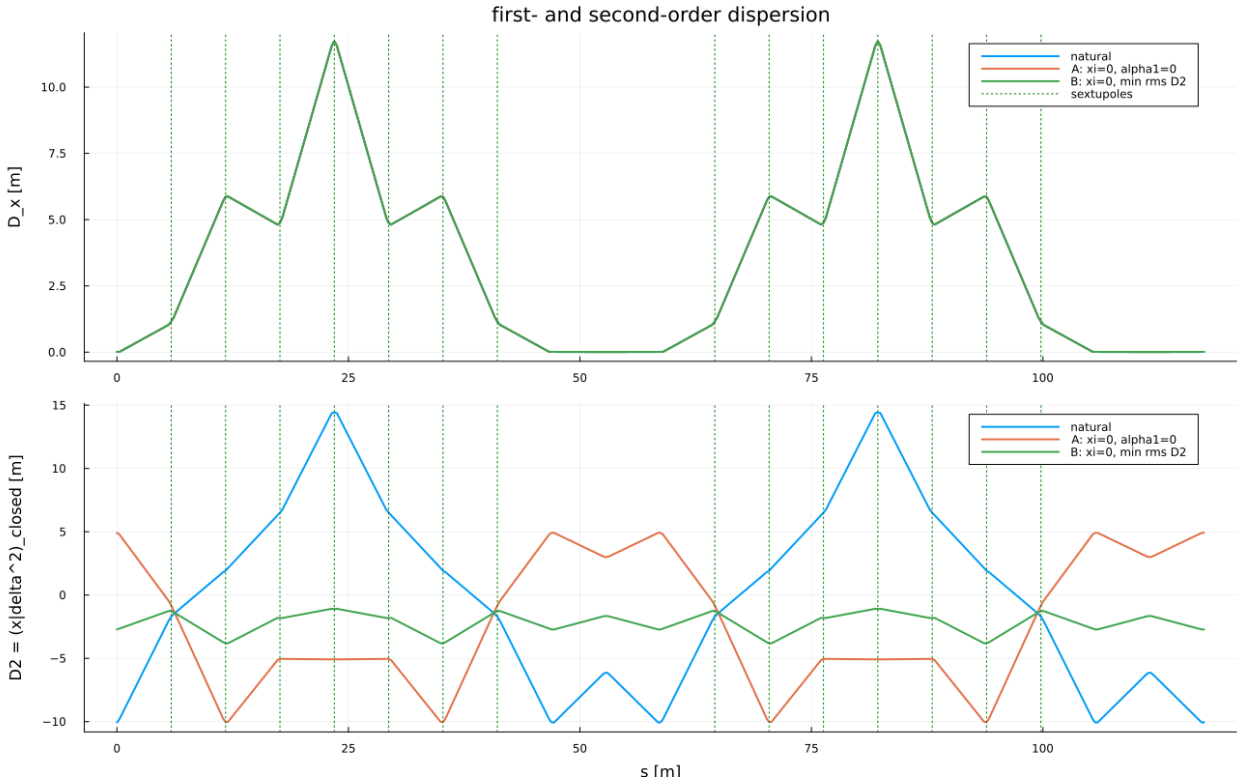


Figure 2: Second-order dispersion along the ring.

6 Chromaticity

At fixed δ the transverse motion about $z_\perp(\delta)$ is linearised. The tunes $Q_{x,y}(\delta)$ from $\cos 2\pi Q = \frac{1}{2}\mathrm{Tr}J(\delta)$ expand as

$$Q(\delta) = Q_0 + \xi_1\delta + \tfrac{1}{2}\xi_2\delta^2.$$

Since δ is conserved without RF, these are invariants of the one-turn map. Cross-check: integrate the exact Hamiltonians at several δ and fit the tune of a small betatron oscillation.

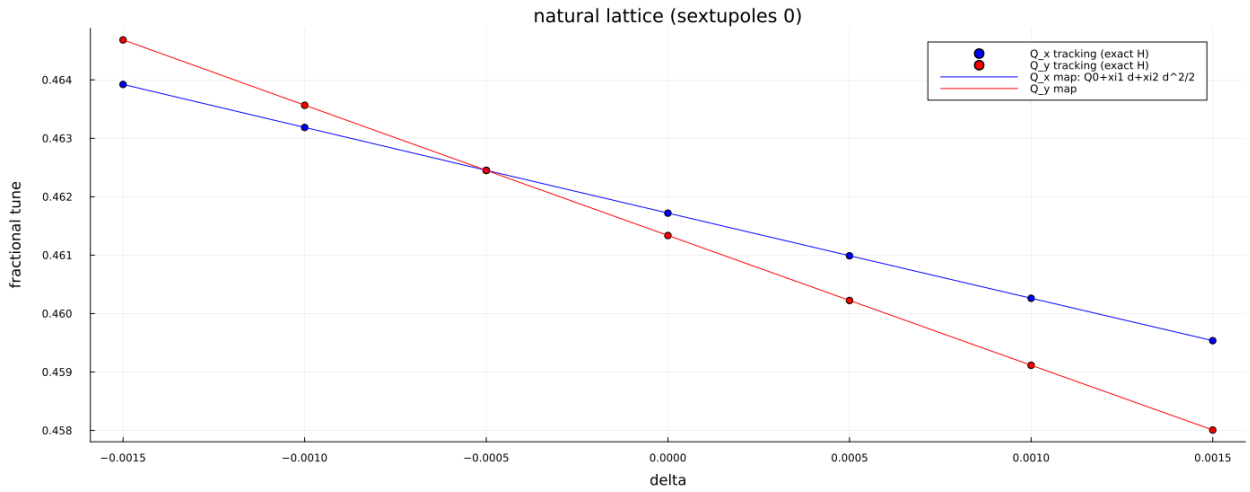


Figure 3: Fractional tunes versus momentum deviation.

7 Finite-length RF cavity

The cavity of length L carries a purely longitudinal on-axis field. With $qA_s/p_0 = -A\cos(\omega t + \psi)$ and $t = s/v_0 - l/v(\delta)$,

$$H_{\mathrm{cav}} = e(\delta) - p_s + A\cos\Phi, \qquad \Phi = \frac{\omega(s - L/2)}{v_0} - \frac{\omega\, l}{v(\delta)} - \phi_s,$$

$$A = \frac{ZV}{p_0 c}\,\frac{c}{\omega L}, \qquad \omega = h\omega_0 = \frac{2\pi h v_0}{C}.$$

Hamilton's equation $\mathrm{d}\delta/\mathrm{d}s = -\partial H/\partial l$ is equivalent to $\mathrm{d}E/\mathrm{d}s = -(ZV/L)\sin\Phi$. Integrated over the cavity the energy gain is

$$\Delta E = ZVT\sin(\phi_s + \phi), \qquad T = \frac{\sin(\theta/2)}{\theta/2}, \quad \theta = \frac{2\pi h L}{C}.$$

H_{cav} depends on s; it is integrated by a symplectic s-dependent scheme (implicit midpoint, Yoshida 4th order). Transverse RF fields are omitted. The thin-kick limit $L \to 0$ recovers an instantaneous energy change $ZV\sin(\phi_s + \phi)$ at fixed (x, p_x, y, p_y, l).

8 Acceleration and adiabatic damping

After the cavity the synchronous particle has $\delta_s \neq 0$. It is made the new reference:

$$p_0' = p_0(1+\delta_s), \quad 1+\delta' = \frac{1+\delta}{1+\delta_s}, \quad p_x' = \frac{p_x}{1+\delta_s}, \quad p_y' = \frac{p_y}{1+\delta_s}, \quad \Delta t' = \Delta t - \Delta t_s.$$

Positions are unchanged. Magnets are assumed to ramp with p_0 (normalised strengths fixed); the RF frequency follows $h\omega_0(p_0)$.

The physical transverse momentum $P_x = p_0 p_x$ is invariant under a longitudinal field, so $p_x \propto 1/p_0$. The Courant–Snyder action in normalised variables then scales as $J \propto 1/p_0$, and betatron amplitudes damp as

$$\frac{A(n)}{A(0)} = \sqrt{\frac{p_{\mathrm{inj}}}{p_0(n)}}.$$

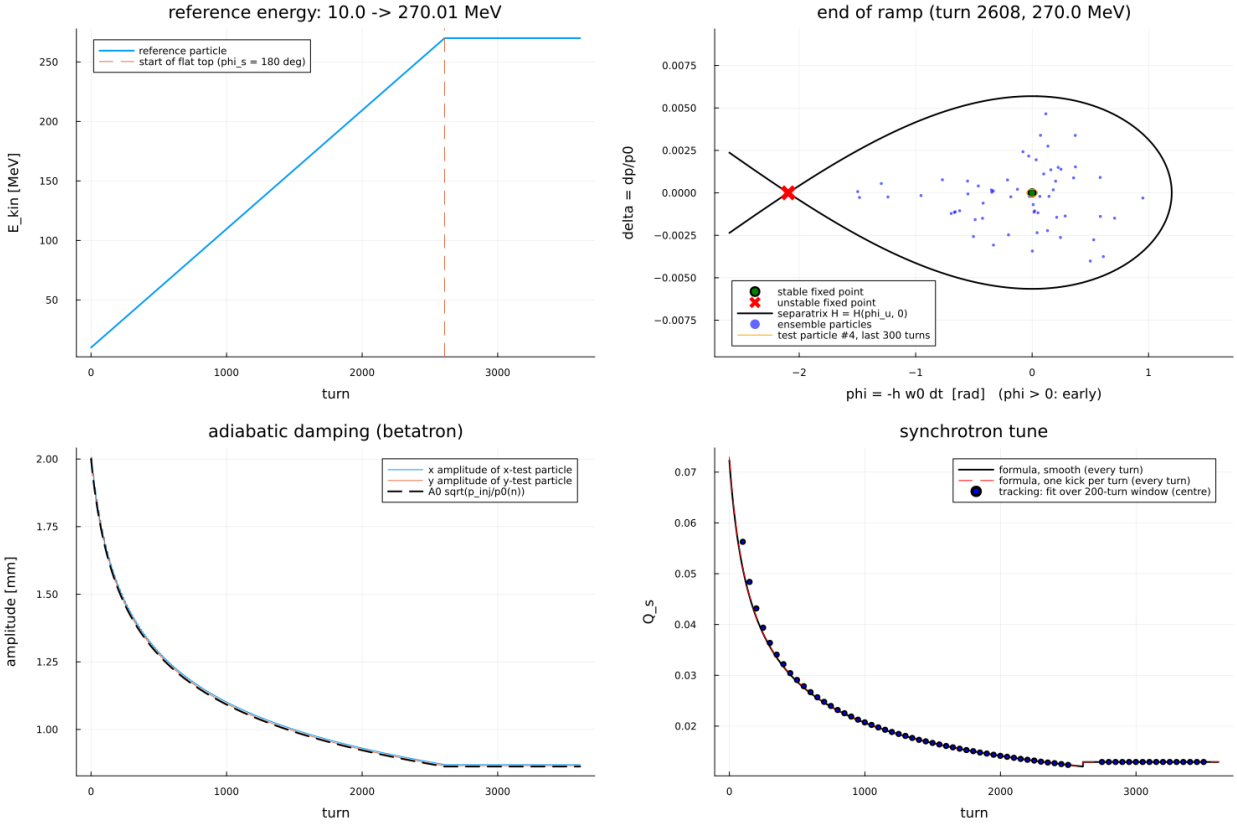


Figure 4: Reference energy, longitudinal phase space with separatrix, adiabatic damping of betatron amplitudes, and synchrotron tune during a ramp.

9 Longitudinal motion from the ring average

9.1 From the 6D Hamiltonian to a 2D synchrotron Hamiltonian

Over one turn the magnetic lattice advances (l, δ) through the exact element Hamiltonians (or, equivalently, through the BLonD `exact` drift built from $\alpha_{0,1,2}$). The RF cavity changes δ at essentially fixed l. Coarse-graining to one map per turn in the variables (ϕ, δ) with $\phi = -h\omega_0\Delta t$ gives

$$\phi_{n+1} = \phi_n - 2\pi h\big(\eta_0\delta_{n+1} + \eta_1\delta_{n+1}^2 + \cdots\big),$$
$$\delta_{n+1} = \delta_n + A\big(\sin(\phi_s + \phi_n) - \sin\phi_s\big), \qquad A = \frac{ZVT}{\beta_0^2 E_0}.$$

Treating the turn number as continuous yields the smooth Hamiltonian

$$H(\phi, \delta) = -2\pi h\Big(\frac{\eta_0\delta^2}{2} + \frac{\eta_1\delta^3}{3}\Big) + A\big(\cos(\phi_s + \phi) + \phi\sin\phi_s\big),$$

which is the average of the full 6D dynamics over one revolution under the assumptions of a single RF station, no transverse–longitudinal coupling in the averaged equations, and δ small enough that the closed-orbit expansion of the path length converges. Fixed points on $\delta = 0$: stable $\phi = 0$, unstable $\phi_u = \pi - 2\phi_s$. The **separatrix** is the level set $H = H(\phi_u, 0)$; the RF bucket is the region around the origin inside it. Stability requires $\eta_0\cos\phi_s > 0$.

Small-amplitude synchrotron tune:

$$Q_s = \sqrt{\frac{h\,\eta_0\,ZVT\cos\phi_s}{2\pi\beta_0^2 E_0}} \qquad \text{(smooth)}, \qquad \cos(2\pi Q_s^{\text{kick}}) = 1 - \tfrac{1}{2}(2\pi Q_s)^2 \qquad \text{(one kick/turn)}.$$

9.2 Element-by-element view of one turn

Left: absolute (ϕ, δ) with the analytic separatrix. Right: change relative to the turn start, with fixed axes set by lattice/RF scales —

- horizontal: $\pm 1.5\,|2\pi h\eta_0|\,\delta_{\text{ens}}$ (full-turn phase-slip scale for the largest $|\delta|$ in the ensemble); - vertical: $\pm 1.5\,A$ (RF kick amplitude).

Only the cavity changes δ within a turn; ϕ changes continuously through path length and velocity.

Animation (see HTML or GIF file): Ensemble during acceleration (left: absolute; right: δ in units of the current bucket half-height).

Animation (see HTML or GIF file): One turn element by element (left: absolute with separatrix; right: intra-turn change on fixed physical axes).

10 Sextupole families

A thin sextupole of integrated strength $k_2 L$ contributes to first-order chromaticity

$$\Delta\xi_{x,y} = \pm\frac{1}{4\pi}\sum k_2 L\,\beta_{x,y} D$$

and to second-order dispersion / α_1 through terms $\propto k_2 L D^2$. Three families placed at maxima of D_x, β_x and β_y therefore control $(\xi_x, \xi_y, \alpha_1, D_2)$ nearly independently. Strengths may be chosen to null ξ_x, ξ_y and optionally α_1 or rms D_2; that choice is an optimisation problem, not part of the lattice model. The natural lattice has vanishing sextupole strengths.

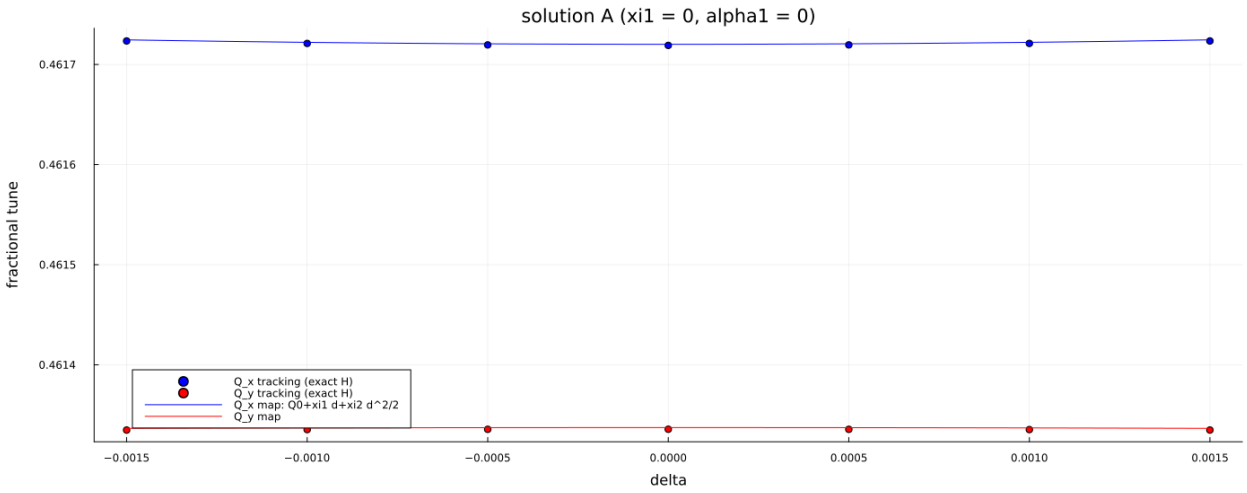


Figure 5: Tunes versus δ after chromaticity correction.

11 Equilibrium momentum shift

In a stationary RF bucket the phase returns to itself after a synchrotron period only if the mean time of flight matches the RF period. With a nonlinear slip factor this condition fixes a nonzero mean relative momentum

$$\delta_{\mathrm{eq}} = -\frac{\eta_1}{\eta_0}\,\frac{\delta_m^2}{2} - \frac{\Lambda}{\eta_0}, \qquad \Lambda = -\frac{2\pi}{C}\big(J_x\,\xi_x + J_y\,\xi_y\big),$$

where `delta_m` is the synchrotron half-amplitude, `eta_0` and `eta_1` are the leading slip-factor coefficients, `xi_x,y` the linear chromaticities and `J_x,y` the Courant-Snyder actions. A cubic truncation of the RF waveform (`sin phi ~ phi - phi^3/6`) multiplies the `eta_1 delta_m^2/2` piece by `1 - phi_hat^2/32` and shifts the synchrotron frequency by `Omega/omega_s = 1 - phi_hat^2/16`.

The mean energy and Lorentz factor receive both the shift `delta_eq` and a kinematic term from the mean square of delta:

$$\Delta E_{\mathrm{eq}} = \beta_s^2 E_s\Big(\delta_{\mathrm{eq}} + \frac{\delta_m^2}{4\gamma_s^2}\Big).$$

11.1 Numeric check (magnetic ring, deuteron 270 MeV, stationary bucket)

Full-ring symplectic tracking in canonical variables, harmonic least-squares extraction of `delta_eq` (constant term of a two-harmonic fit of delta over many synchrotron periods; half-split systematic error at most about 1e-8). Phase reported in the late-positive convention of the analytic note.

Longitudinal only (Lambda = 0), delta_m scan:

delta_m	**delta_eq (track)**	**-eta1 dm^2/(2 eta0)**	**rel.**
5.0e-4	3.45e-7	2.38e-7	0.45
1.0e-3	1.06e-6	9.50e-7	0.11
2.0e-3	3.88e-6	3.78e-6	0.025

Betatron contribution. At `delta_m ~ 3e-4` and vertical amplitude 20 mm, `Lambda_fit / Lambda_form = 1.009`; with sextupoles set to `xi ~ 0` the fitted Lambda drops by an order of magnitude toward the longitudinal baseline.

Cubic phase. For `phi_hat` in [0.30, 0.50], the measured `Omega/omega_s` agrees with `1 - phi_hat^2/16` to 1e-4, and `<(delta-deq)^2> / (dm^2/2)` stays within 1% of unity (cubic correction `1 - phi_hat^2/32` is a few parts in 1e3).

These relations are verified by symplectic tracking of the full ring Hamiltonians against the formulae above.

12 Assumptions

Remaining modelling assumptions (accepted design choices such as injection energy, RF parameters, CODATA mass, and the interpretation of the lattice file are recorded in the Programmer Manual, not here).

1. Hard-edge magnets, no fringe fields.
2. No space charge, impedances, radiation or alignment errors.
3. RF field purely longitudinal on axis (no transverse RF).
4. Magnets and RF frequency scale with the reference momentum during

acceleration.

1. Single RF station per turn for the averaged longitudinal Hamiltonian.
2. Truncated maps are symplectic only up to the truncation order; exact

integration of H is symplectic to round-off.